Graphing Inequalities Error Analysis

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

1. Solid or dashed?

ORIGINAL PROBLEM

Graph $y > 2x - 1$

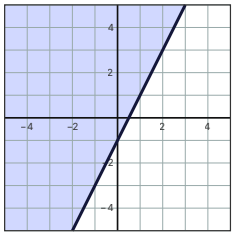
INCORRECT WORKED SOLUTION

$y > 2x - 1$

Slope 2, y-intercept −1.

Draw a SOLID line.

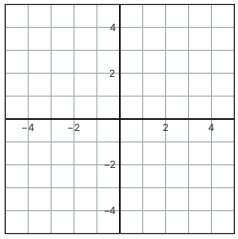
Shade above the line.



A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)



2. Solid or dashed?

ORIGINAL PROBLEM

Graph $y \leq -x + 3$

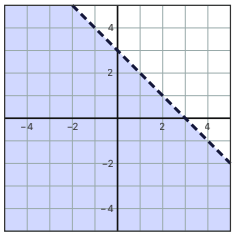
INCORRECT WORKED SOLUTION

$y \leq -x + 3$

Slope −1, y-intercept 3.

Draw a DASHED line.

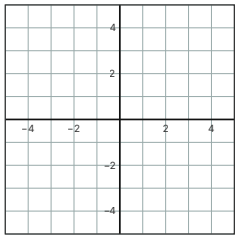
Shade below the line.



A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)



Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

Graphing Inequalities Error Analysis

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3. Which side to shade

ORIGINAL PROBLEM

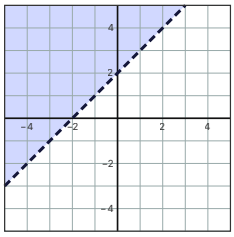
Graph $y < x + 2$

INCORRECT WORKED SOLUTION

$y < x + 2$

Dashed line through (0, 2) with slope 1.

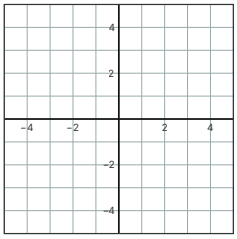
Shade above the line.



A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)



4. Vertical boundary

ORIGINAL PROBLEM

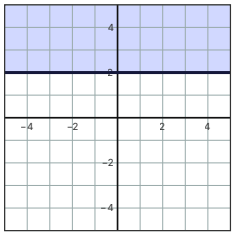
Graph $x \geq 2$

INCORRECT WORKED SOLUTION

$x \geq 2$

Draw a solid horizontal line through $y = 2$.

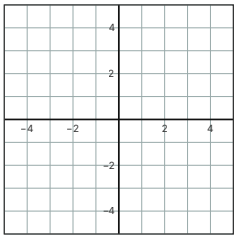
Shade above the line.



A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)



Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

Graphing Inequalities Error Analysis

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

5. Plotting the slope

ORIGINAL PROBLEM

Graph $y \geq \frac{1}{2}x - 3$

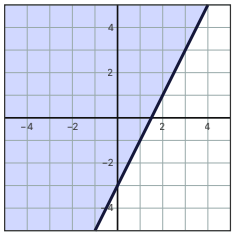
INCORRECT WORKED SOLUTION

$y \geq \frac{1}{2}x - 3$

Start at $(0, -3)$.

Slope $\frac{1}{2}$: go up 2, right 1 to plot $(1, -1)$.

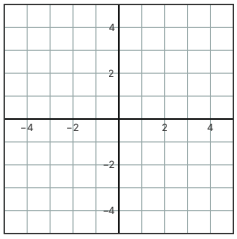
Solid line, shade above.



A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)



6. Reading the y-intercept

ORIGINAL PROBLEM

Graph $y < -2x + 4$

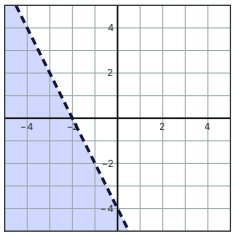
INCORRECT WORKED SOLUTION

$y < -2x + 4$

Start at $(0, -4)$.

Slope -2: down 2, right 1.

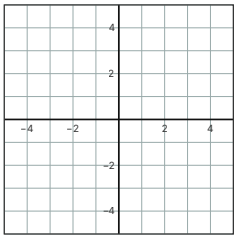
Dashed line, shade below.



A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)



Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

Graphing Inequalities Error Analysis

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

7. Rewriting first

ORIGINAL PROBLEM

Graph $2x + y \leq 4$

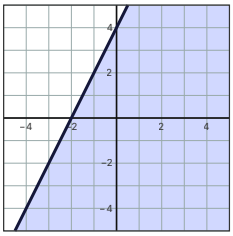
INCORRECT WORKED SOLUTION

$2x + y \leq 4$

$y \leq 4 + 2x$

$y \leq 2x + 4$

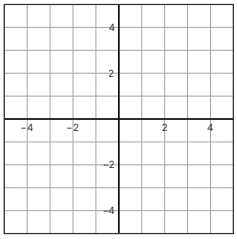
Solid line through (0, 4), slope 2; shade below.



A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)



8. Using a test point

ORIGINAL PROBLEM

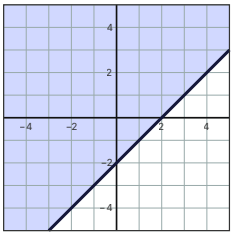
Graph $y \leq x - 2$

INCORRECT WORKED SOLUTION

$y \leq x - 2$

Test (0, 0): $0 \leq -2$ is false.

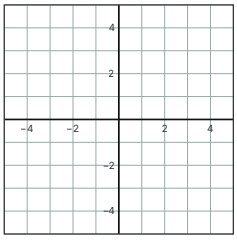
Shade the side that contains (0, 0).



A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)



Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

Graphing Inequalities Error Analysis

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

9. Horizontal boundary

ORIGINAL PROBLEM

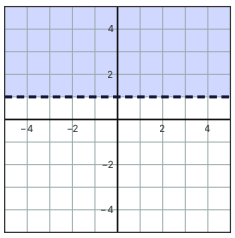
Graph $y > -1$

INCORRECT WORKED SOLUTION

$y > -1$

Draw a dashed horizontal line through 1 on the y-axis.

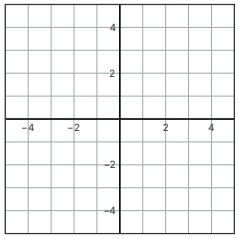
Shade above the line.



A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)



10. Bad test point

ORIGINAL PROBLEM

Graph $y > x$

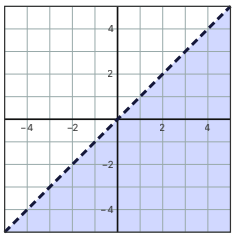
INCORRECT WORKED SOLUTION

$y > x$

Dashed line through the origin with slope 1.

Test (1, 1): $1 > 1$ is false.

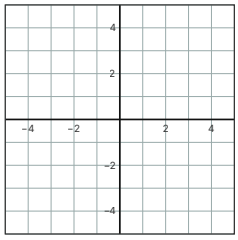
Shade below the line.



A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)



Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points